Lab Tutorial: Feature Engineering

Objective

Investigate and address the issue of a negative weight for the "Number of Bedrooms" feature in multivariate linear regression when predicting house prices.

Possible Causes

1. **Multicollinearity with Size**:
    - The number of bedrooms and size are likely correlated. Larger houses tend to have more bedrooms, but if size already explains most of the variation in price, adding the number of bedrooms might introduce redundancy.
    - The model adjusts for this by assigning a negative weight to balance the overlapping information.
2. **Nonlinear Relationships**:

    The relationship between price and bedrooms might not be strictly linear. For instance, an increase in bedrooms might not always correlate with a proportional increase in price.

3. **Relative Importance**:

    Size (square footage) might be the dominant factor influencing price, and the number of bedrooms might have a minor or inverse effect when size is already accounted for.

4. **Data Distribution**:

    If the dataset contains houses with the same number of bedrooms but vastly different sizes, the number of bedrooms may appear to have a weaker or inverse impact on price.

Suggested Actions

1. Plot a histogram to identify skewness or anomalies in the data

```
import matplotlib.pyplot as plt
plt.hist(data[:, 1], bins=10)  # Column 1: Number of Bedrooms
plt.title("Distribution of Number of Bedrooms")
plt.xlabel("Number of Bedrooms")
plt.ylabel("Frequency")
```

```
plt.show()
```

2. Data Visualization

Plot price vs. bedrooms and price vs. size to visually inspect the relationships.

There is a clear positive relationship between house size (square feet) and price. Larger houses tend to have higher prices.

Prices for houses with the same number of bedrooms can vary significantly, which suggests that the number of bedrooms alone may not be a strong predictor of price.

```
import matplotlib.pyplot as plt

# Extract columns
size = data[:, 0]
bedrooms = data[:, 1]
price = data[:, 2]

# Visualize the relationships
plt.figure(figsize=(12, 5))

# Plot Size vs Price
plt.subplot(1, 2, 1)
plt.scatter(size, price, color='blue')
plt.title('Size vs Price')
plt.xlabel('Size (Square Feet)')
plt.ylabel('Price (USD)')


# Plot Bedrooms vs Price
plt.subplot(1, 2, 2)
plt.scatter(bedrooms, price, color='green')
plt.title('Bedrooms vs Price')
plt.xlabel('Number of Bedrooms')
plt.ylabel('Price (USD)')

# Show plots
plt.show()
```

3. Correlation Analysis

Compute the correlation matrix to see the relationships between features.

Code for correlation matrix

Compute the correlation between "Number of Bedrooms" and "Price".

```
#Check the correlation matrix to identify multicollinearity.
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset
# Replace 'lab2data.txt' with the actual file path if needed
data = np.loadtxt('lab2data.txt', delimiter=',')

# Extract columns
size = data[:, 0]
bedrooms = data[:, 1]
price = data[:, 2]

# Combine data into a single array for correlation analysis
data_combined = np.column_stack((size, bedrooms, price))

# Calculate the correlation matrix
correlation_matrix = np.corrcoef(data_combined, rowvar=False)

# Plot the correlation matrix
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, fmt=".2f", cmap="coolwarm",
xticklabels=['Size', 'Bedrooms', 'Price'], yticklabels=['Size', 'Bedrooms', 'Price'])
plt.title('Correlation Matrix')
plt.show()
```

The correlation between size and price is very strong (0.85), indicating that size is a significant predictor of price.

The correlation between bedrooms and price is weaker (0.44) compared to size, suggesting that the number of bedrooms is a less significant factor for predicting price.

The correlation between bedrooms and size is moderate (0.56), indicating some overlap between these two features.

The moderate correlation between bedrooms and size suggests **multicollinearity**, which might lead the regression model to assign a negative weight to bedrooms as it compensates for the influence already captured by size.

Recommendations

1. **Feature Selection or Transformation**:

Instead of using both size and bedrooms directly, you could create a derived feature like size/bedrooms.

2. **Regularization**:

Use Ridge or Lasso regression to mitigate the effects of multicollinearity and stabilize the coefficients.